

ONE8 PRODUCT PAGE A/B TEST

(Final Outputs —Report)

1. Executive Summary:

One8 tested an enhanced product page (**Variant B** — richer product information, customer reviews, and size guidance) against the existing page (**Variant A**) across 10,000 website sessions.

Variant B produced a statistically significant lift in both add-to-cart rate (+11.2%) and purchase conversion (+15.9%), with a directional but not-yet-significant lift in revenue per session (+12.6%).

Recommendation: Ship Variant Prioritizing mobile users and the Formal Footwear category, where the effect was strongest — while extending the test to confirm the revenue impact.

Metric	Variant A	Variant B	Result
Purchase conversion	10.96%	12.70%	+15.9%, p=0.0068
Add to cart rate	26.06%	28.98%	+11.2%, p=0.0011
Revenue per session	377.42Rs	425.08Rs	+12.6%, p=.1027

2. Business Problem:

One8 wants to determine whether a new product-page experience (Variant B) improves customer conversion and revenue compared with the existing experience (Variant A). The current product page offers minimal information — customers frequently abandon the page without adding items to cart, particularly for products where fit, sizing, or material quality are a purchase concern (footwear, apparel).

*The question this project answers: **Is a richer, more informative product page worth the investment to build and roll out?***

3. Experiment Design

- *Type: Randomized controlled A/B test, single-factor (page variant)*
- *Control (**Variant A**): existing, minimal product page*
 - *Treatment (**Variant B**): enhanced page — expanded product information, customer reviews, and size guidance*
- *Assignment: every session randomly and independently assigned to A or B (~50/50 split), eliminating selection bias*
- *Primary metrics: purchase conversion rate, add-to-cart rate* • *Secondary metric: revenue per session*
- *Segmentation cuts: device type, product category*

4. Dataset

The dataset is synthetically generated for this portfolio project. No real One8 customer data was used. Product categories and price bands are grounded in One8's actual, publicly listed product lines (One8 x Puma athleisure/performance wear, One8 Select formal footwear, One8 Fragrances) rather than generic placeholder categories, so the simulation reflects a realistic assortment.

Property	Detail
Sessions	10,000(4,993 in A & 5,007 in B)
Assignment	Random A/B split, -50/50
Product Categories	Athleisure Apparel, Performance Footwear, Formal Footwear, Fragrances, Accessories
Devices	Mobile, Desktop, Tablet
Fields captured	Price, device, variant, add-to-cart flag, purchase flag, revenue

5.SQL Analysis:

All funnel metrics were computed directly against the session-level data using SQL aggregation — the same approach used on a live analytics warehouse:

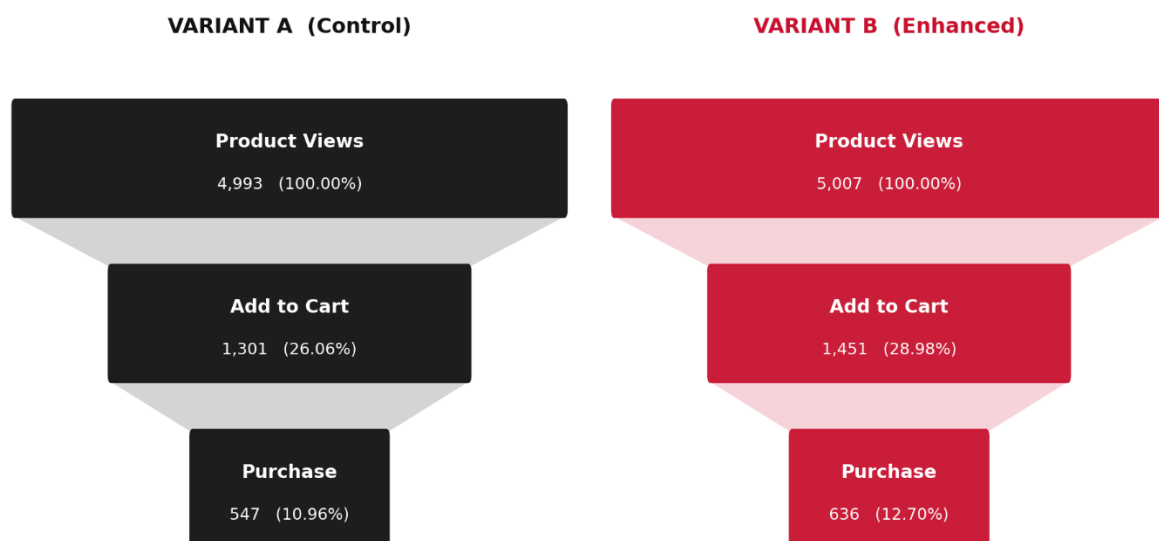
```
SELECT variant, COUNT(*) AS sessions, SUM(added_to_cart) AS  
add_to_carts, SUM(purchased) AS purchases,  
ROUND(100.0 * SUM(purchased) / COUNT(*), 2) AS  
conversion_rate_pct FROM sessions GROUP BY variant;
```

This produced the funnel counts (views, add-to-carts, purchases) and the contingency table used for significance testing, plus device- and category-level breakdowns via the same GROUP BY pattern.

6. Conversion Analysis:

Both funnel stages show a clear, consistent gap in favour of Variant B. The gap widens at each stage — Variant B doesn't just attract more initial interest, it converts that interest into purchases more effectively than Variant A.

CUSTOMER FUNNEL — VARIANT A vs VARIANT B



7. Statistical Testing:

Observing a higher rate for Variant B isn't proof of a real effect — it could be random chance. Each metric was tested formally:

Test applied	Metric	p-value	Significant 95% ?
Two proportion z-test	Add to cart rate	0.0011	Yes
Two proportion z-test	Purchase conversion	0.0068	Yes
Welch's t-test	Revenue per session	0.1027	No

A p-value below 0.05 means the observed gap is very unlikely to be due to random chance. Both funnel-stage metrics clear this bar comfortably; the revenue improvement is promising but not yet statistically confirmed with this sample size.

8. Revenue Analysis:

Revenue is a noisier metric than a conversion rate — it's driven by a relatively small number of high-value transactions, so it naturally requires a larger sample (or longer test duration) to reach statistical confidence.

The direction is positive and consistent with the conversion gains, but should be treated as preliminary, not confirmed, until validated on more data.

Metric	Variant A	Variant B	Difference
Revenue per session	377.42	425.08	+47.66(+12.6%)

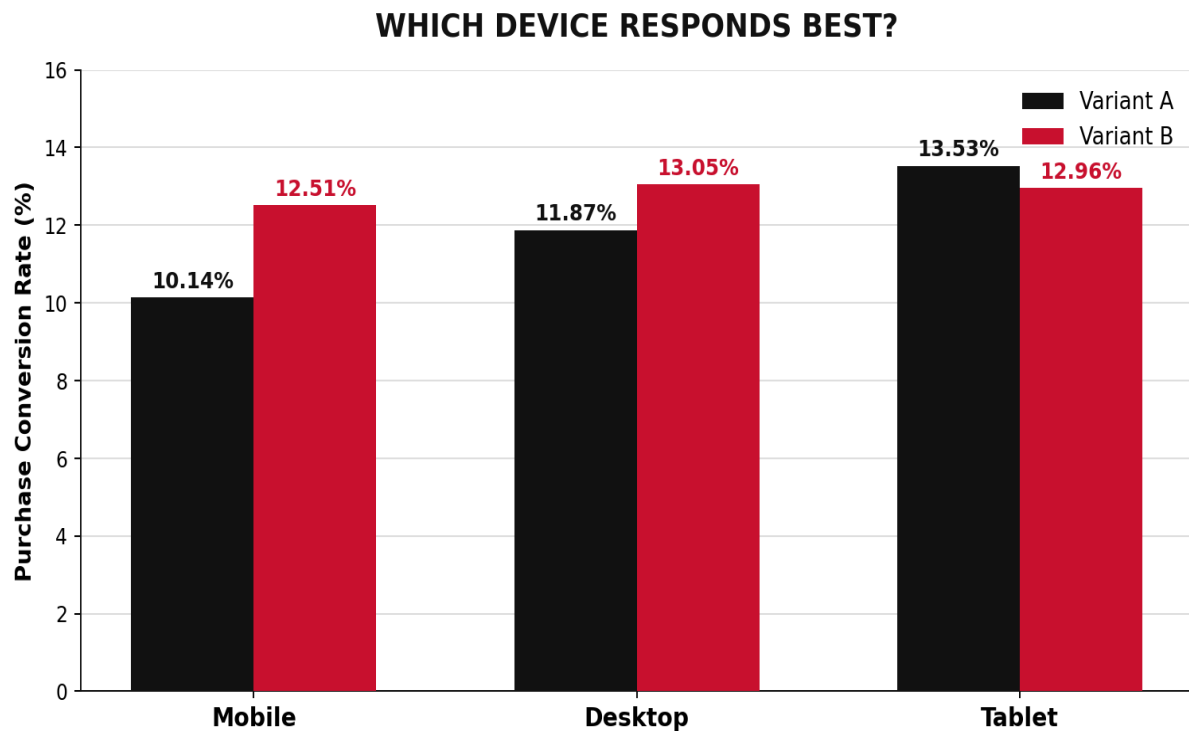
9. Key Findings:

- Mobile users respond most strongly to Variant B(+2.37points), consistent with the idea that additional product detail compensates for the reduced screen space and information density of mobile browsing.
- Formal Footwear shows the largest category lift (+4.55 points) — a higher consideration, higher-price purchase where sizing and material detail meaningfully reduce hesitation.
- Fragrances show almost no lift(+0.16points)—a lower-price, lower-risk purchase needs less convincing, so extra product information adds little value there.
- Tablet showed a small negative shift (−0.57 points); given tablet is only ~8% of total traffic, this is most likely sampling noise rather than a genuine negative effect, and should be monitored rather than acted on.

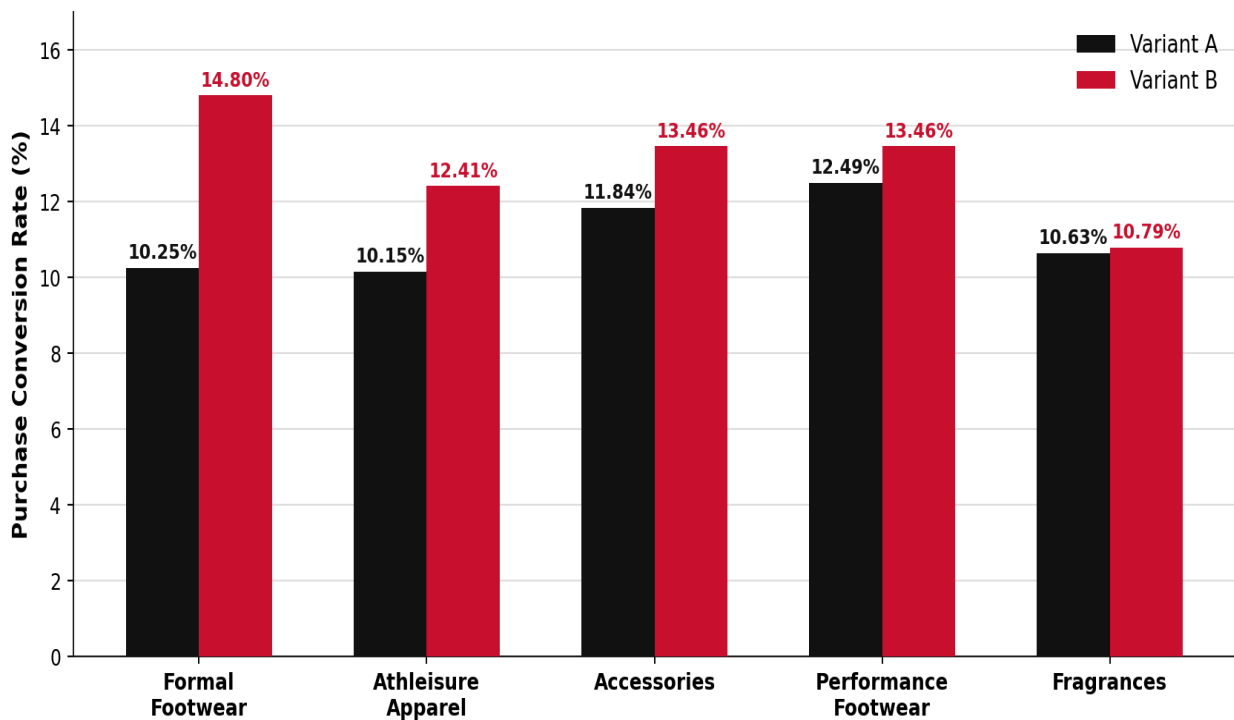
10. Business Recommendation:

1. Ship Variant B site-wide — the conversion and add-to-cart gains are statistically significant and consistent across segments.

2. Prioritize rollout on mobile and on Formal Footwear category pages first, where the impact is largest and the investment pays back fastest.



WHICH PRODUCT CATEGORY RESPONDS BEST?



3. Continue monitoring revenue per session over a longer window before including a firm revenue projection in any business case — the current result is directionally positive but not yet statistically confirmed.

4. Retest tablet specifically once more traffic accumulates, given the small and inconclusive negative signal there.

11. Limitations

- **Synthetic data:** this analysis uses a simulated dataset, not live One8 traffic. Real customer behaviour may differ in ways this simulation cannot capture (seasonality, marketing campaign overlap, returning vs. new customer mix).
- **Revenue sample size:** the revenue metric did not reach statistical significance at this sample size; a larger or

longer test is needed before treating the revenue lift as confirmed.

- Single test window: results reflect a 30-day simulated window; real-world tests should be checked for weekday/weekend and seasonal effects before a permanent rollout decision.
- No qualitative signal: this analysis captures what users did, not why — a complementary user research or session-recording review would strengthen the case before a full rollout.